Methods

Distinctiveness

- In each bigram, phone2's **distinctiveness** w/ phone1 was calculated as the mean Euclidean distance between their **articulatory/acoustic matrices**.
- The matrices were extracted through self-supervised autoencoders.
- Acoustic/articulatory data → encoder → latent space → decoder → reconstructed data

Methods

Distinctiveness

- In each bigram, phone2's **distinctiveness** w/ phone1 was calculated as the mean Euclidean distance between their **articulatory/acoustic matrices**.
- Acoustic matrices: the phone token's MFCCs (30 coefficients; 10 frames)
- Articulatory matrices (EMA/annotated MRI):
    - The temporal dimension of the video clip for each token was normalized to 5 frames (AE-MRI-annotated) & 10 frames (AE-EMA).
- MFCCs/x, y, (z) **coordinates** → 1D CNN + temporal pooling + MLP → <u>1024D latent matrix</u>